

Module 09: TCP Congestion Detection: Fast Retransmit and Fast Recovery

Unit 4: Transport Layer

Loss Recovery & Reno Dynamics

Congestion Signals Taxonomy, Severe RTO Timeout vs Mild 3 Duplicate ACKs, TCP Tahoe vs Reno Architectural Comparison, and Fast Recovery State Machine.

TCP Congestion Detection: Timeout vs Fast Retransmit & Fast Recovery

1. Severe Congestion: RTO Timer Expiration

Network Diagnosis: Severe Bottleneck!

- Multiple packets dropped; intermediate router queues fully choked.
- No duplicate ACKs can return to the sender.

Action: Multiplicative decrease to rock bottom!

Mandatory Reaction Steps:

$ssthresh = cwnd / 2$

$cwnd = 1 \text{ MSS}$ (Reset to ground zero)

Resets to Slow Start mode (Exponential increase from 1 MSS).

High throughput penalty — Takes multiple RTTs to recover speed!

TCP Tahoe Philosophy (RFC 1122):

Treats ALL packet losses identically (whether Timeout or Dup ACKs).

Always cuts $ssthresh = cwnd / 2$, drops $cwnd = 1 \text{ MSS}$, and forces Slow Start! Highly conservative and causes throughput collapse.

2. Mild Congestion: 3 Duplicate ACKs (TCP Reno)

Network Diagnosis: Isolated Packet Loss!

- Subsequent segments ARE reaching receiver (triggering Dup ACKs).
- Network pipeline is still flowing, not completely jammed.

Action: Fast Retransmit + Fast Recovery!

TCP Reno Reaction Steps (RFC 2581):

$ssthresh = cwnd / 2$

$cwnd = ssthresh + 3 \text{ MSS}$ (Fast Recovery)

Resends missing segment immediately (Fast Retransmit).

Skips Slow Start completely! Resumes Congestion Avoidance directly.

Modern TCP Variants (NewReno & SACK):

- **TCP NewReno (RFC 6582):** Handles multiple packet drops in 1 window.
- **TCP SACK (RFC 2018):** Selective Acknowledgment allows receiver to report non-contiguous blocks, retransmitting only lost chunks!

1. Severe vs Mild Congestion Response

RTO Timeout: $ssthresh = cwnd / 2$, $cwnd = 1 \text{ MSS}$ (Full reset to Slow Start)

3 Duplicate ACKs: Fast Retransmit missing segment immediately without waiting for RTO.

TCP Reno Fast Recovery: $ssthresh = cwnd / 2$, $cwnd = ssthresh$ (Resumes Congestion Avoidance directly).

2. Evolution of TCP Loss Recovery

TCP Version	Loss Detection	Recovery Behavior
Tahoe	Timeout or 3 Dup ACKs	Always cuts cwnd to 1 MSS; enters Slow Start
Reno	3 Duplicate ACKs	Fast Retransmit + Fast Recovery; skips Slow Start
NewReno	Partial ACKs	Recovers multiple packet losses per window without RTO
SACK	Selective ACK blocks	Retransmits exclusively missing packet holes